
2a	$g = GM / R^2$ $M = [9.81 \times (6.4 \times 10^6)^2] / 6.67 \times 10^{-11} = 6.02 \times 10^{24}$ $\approx 6.0 \times 10^{24}$ kg	 M1 A0
2b(i)	Gravitational force provides for centripetal force, $GM / r^2 = r\omega^2$ For geostationary satellite, $\omega = 2\pi / (24 \times 3600)$ Hence, $6.67 \times 10^{-11} \times 6.02 \times 10^{24} = r^3 \times \omega^2$ $r = 4.23 \times 10^7 \approx 4.2 \times 10^7$ m	B1 M1 A1

2b(ii)	Potential energy at Equator, $\Phi_e = -\dfrac{GMm}{R_e}$ Potential energy at geostationary orbit, $\Phi_o = -\dfrac{GMm}{R_o}$ Increase in potential energy $= \Phi_o - \Phi_e = -\dfrac{GMm}{R_o} - \left(-\dfrac{GMm}{R_e}\right) = GMm\left(\dfrac{1}{R_e} - \dfrac{1}{R_o}\right)$ $= 6.67\times10^{-11}\times6.02\times10^{24}\times650\left(\dfrac{1}{6.4\times10^{6}} - \dfrac{1}{4.23\times10^{7}}\right)$ $= 3.46 \times 10^{10} \approx 3.5 \times 10^{10}$ J	 C1 A1
2c	Total energy of the satellite will decrease (due to work done against resistive forces). Radius will decrease since <u>total energy = -GMm/2r</u>, so the kinetic energy will increase <u>since KE = GMm/2r</u>.	M1 M1 A1

[illegible]	**Specific latent heat of vaporization** is the amount of thermal energy per unit mass	B1